Problem Brief & Calibrated Value Proposition Canvas

Course: User Experience Design & Human-Computer Interaction (ICC612 / ICC716) — UFRO 2026-2
Initiative 03: Assessment Feedback that Students Actually Use
Deliverable: Milestone 1 (Hito 1) — Strategy Layer Artifact
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1. What is a Problem Brief? (Conceptual Definition)

In formal UX Design, a **Problem Brief** is an executive alignment document (belonging to the **Strategy Layer** in Jesse James Garrett's model). It bridges business/academic intent with user research findings.

Instead of jumping straight into interface layout, the Problem Brief establishes:

1. **Core Problem Statement:** What specific breakdown occurs in the user experience, who is affected, and why existing solutions fail.
2. **Target Audience & Roles:** The validated personas involved.
3. **Core Value Proposition:** What transformation our system delivers.
4. **Scope Boundaries (Guardrails):** Explicitly what is inside the scope (In-Scope) and what is strictly excluded (Out-of-Scope) to avoid *feature bloat*.
5. **Success Signals (UX Metrics):** How we prove that the design successfully closed the problem loop.

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2. Project Problem Brief (Group 5 — Initiative 03)

2.1 Background & Context

In tertiary education (specifically in project-based disciplines such as Engineering and Design at Universidad de La Frontera), evaluation feedback arrives late and scattered across disparate formats (annotated PDFs, spreadsheet grading tables, LMS comment threads, and email exchanges). Because rubric feedback is predominantly descriptive rather than prescriptive, students struggle to translate critical comments into concrete modifications for their subsequent submissions.

2.2 Problem Statement

> ***"Students (Valentina & Samon)*** *experience confusion, anxiety, and decision paralysis when receiving assessment results because rubric comments lack actionable guidance and are disconnected from the resubmission workflow. Concurrently,* ***course instructors (Martina)*** *face severe grading fatigue because verifying whether students applied past corrections requires tedious, manual side-by-side document comparisons. Consequently, the formative learning loop remains broken, leading to repeated errors and superficial compliance."*

2.3 Proposed Solution: "LoopFeedback" (Evidence-Bound Feedback System)

A dedicated, lightweight evaluation feedback environment that transforms passive rubric feedback into a closed-loop action workflow:

- **Deconstructs evaluation rubrics** into individual criterion levels with clear deficiency explanations.
- **Pairs weak criteria directly with prescriptive Action Cards** that specify exact corrective guidelines.
- **Enforces Bidirectional Evidence Binding:** Students cannot claim a correction is done without linking concrete evidence (file excerpt, page reference, screenshot, or code link).
- **Streamlines Instructor Verification:** Instructors inspect only the submitted evidence alongside the original comment, eliminating manual document comparisons.

2.4 Scope Boundaries (Guarding Against Scope Creep)

- **In-Scope (Flujo Principal + 2 Flujos Secundarios):**
    1. *Core Flow:* Identify Weak Criterion $\rightarrow$ Access Prescribed Action $ \rightarrow$ Submit & Bind Corrective Evidence.
    2. *Secondary Flow 1:* Instructor Evidence Verification & Status Approval (`Pending` $ \rightarrow$ `Evidence Submitted` $\rightarrow$ `Approved`).
    3. *Secondary Flow 2:* Longitudinal Criterion Comparison between consecutive submissions.
- **Strictly Out-of-Scope (Justified Exclusion):**
    - General academic calendar, course timetables, or class scheduling.
    - Full-fledged institutional LMS gradebook or GPA calculators.
    - Unstructured free-form chat/direct messaging between students and professors. *(These features introduce clutter, obscure the feedback interaction model, and violate the course's strict scope restriction).*

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3. Calibrated Value Proposition Canvas (Covering All 3 Personas)

3.1 Customer Profile (Right Side)

A. Customer Jobs

- **Valentina (Learner):** Understand the root cause of lost points; locate exact rubric expectations; execute verifiable corrections.
- **Samon (Prioritizer):** Triage multiple revision tasks against impending deadlines; correlate visual prototype changes with rubric feedback.
- **Martina (Instructor):** Deliver high-quality formative feedback efficiently; verify student compliance without reading entire documents again.

B. Pains (Frustrations & Obstacles)

- **Valentina:** Vague comments ("elaborate more"); scattered feedback files; anxiety that resubmitted fixes will be overlooked.
- **Samon:** Unranked feedback dumps causing cognitive overload; losing track of which design version received which comment.
- **Martina:** 20+ hours of manual document cross-referencing; dealing with students claiming they "fixed everything" with zero proof.

C. Gains (Desired Outcomes)

- **Valentina:** Immediate clarity on what to fix; confidence that evidence is formally registered and visible to the grader.
- **Samon:** Ranked action backlog sorted by rubric weight; clear version association for design assets.
- **Martina:** 60% reduction in verification time for second deliverables; auditable proof of student learning evolution.

3.2 Value Map (Left Side — Solution Alignment)

A. Products & Core Capabilities

1. **Interactive Rubric Breakdown:** Dynamic display linking overall score to discrete criterion levels and rationale.
2. **Prescriptive Action Cards:** Standardized action prompts associated with penalized rubric criteria.
3. **Evidence-Binding Input Module:** Upload and deep-linking interface specifically for corrective evidence.
4. **Verification Queue (Instructor View):** Side-by-side modal displaying: *Original Comment* $\leftrightarrow$ *Student Linked Evidence* $\leftrightarrow$ *Approval Button*.

B. Pain Relievers

- **For Valentina:** Converts abstract critique into concrete, checkable instructions with explicit success criteria.
- **For Samon:** Sorts action cards by rubric point value and deadline; attaches version tags so revisions remain contextualized.
- **For Martina:** Completely eliminates full-document comparative reading; exposes the student's exact evidence in a single click.

C. Gain Creators

- **Closes the Formative Loop:** Provides concrete, objective proof of whether learning occurred between submission 1 and 2.
- **Objective Assessment Transparency:** Removes perceived grading subjectivity by maintaining a single source of truth for criteria and evidence.
- **Effort-Optimized Iterations:** Focuses student revision energy exclusively where the rubric penalty occurred.

3.3 Persona-to-Canvas Mapping Summary

Canvas Dimension	Valentina Rojas	Samon Baker	Martina Soto
Addressed Job	Understand and correct weak criteria.	Prioritize fixes within limited time.	Rapidly verify student corrections.
Relieved Pain	Ambiguous comments & scattered files.	Overwhelming unranked critique.	Tedious side-by-side re-grading.
Created Gain	Verifiable submission recognized by teacher.	Organized, weighted action backlog.	Fast, auditable evidence inspection.
Core UI Feature	Action Card + Evidence Uploader.	Priority / Weight Filter & Version Tag.	Verification & Comparison Queue.